The force of attraction or repulsion between two magnetic poles is directly proportional to the product of their pole strengths and inversely proportional to the square of the distance between them.

Mathematically, we can write

$$\vec{F} \propto \frac{q_{m_A} q_{m_B}}{r^2} \hat{r}$$

where m_A and m_B are pole strengths of two poles and r is the distance between two magnetic poles.

$$\vec{F} = k \frac{q_{m_A} q_{m_B}}{r^2} \hat{r} \qquad (3.7)$$

In magnitude, $$\vec{F} = k \frac{q_{m_A} q_{m_B}}{r^2} \qquad (3.8)$$

where k is a proportionality constant whose value depends on the surrounding medium. In S.I. unit, the value of k for free space is $k = \frac{\mu_\circ}{4\pi} \approx 10^{-7} H m^{-1}$, where μ_o is the absolute permeability of free space (air or vacuum).

EXAMPLE 3.5

The repulsive force between two magnetic poles in air is 9 x 10^{-3} N. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole.

Solution:

The force between two poles are given by

$$\vec{F} = k \frac{q_{m_A} q_{m_B}}{r^2} \hat{r}$$

The magnitude of the force is

$$F = k \frac{q_{m_A} q_{m_B}}{r^2}$$

Given : F = 9 x 10^{-3}N, r = 10 cm = 10 x 10^{-2} m

Therefore,

$$9 \times 10^{-3} = 10^{-7} \times \frac{q_m^2}{\left(10 \times 10^{-2}\right)^2} \Rightarrow q_m = 30 N T^{-1}$$

3.2.1 Magnetic field at a point along the axial line of the magnetic dipole (bar magnet)

Consider a bar magnet NS as shown in Figure 3.16. Let N be the North Pole and S be the south pole of the bar magnet, each of pole strength q_m and separated by a distance of 2*l*. The magnetic field at a point C (lies along the axis of the magnet) at a distance from the geometrical center O of the bar magnet can be computed by keeping unit north pole (q_{mc} = 1 A m) at C. The force experienced by the unit north pole at C due to pole strength can be computed using Coulomb's law of magnetism as follows:

The force of repulsion between north pole of the bar magnet and unit north pole at point C (in free space) is

$$\vec{F}_N = \frac{\mu_\circ}{4\pi} \frac{q_m}{(r-l)^2} \hat{i} \qquad (3.9)$$

where r – *l* is the distance between north pole of the bar magnet and unit north pole at C.

The force of attraction between South Pole of the bar magnet and unit North Pole at point C (in free space) is

$$\vec{F}_S = -\frac{\mu_\circ}{4\pi} \frac{q_m}{(r+l)^2} \hat{i} \qquad (3.10)$$

where r + *l* is the distance between south pole of the bar magnet and unit north pole at C.

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